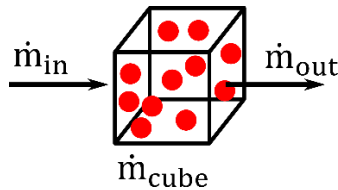


Conservation Laws:

- Describe the fluid flow behavior
- To predict the flow behavior and flows, we apply rules the flow must follow.
 - Conservation of mass
 - Conservation of momentum
 - Conservation of energy

Conservation of Mass (a.k.a Continuity Equation):

Mass can neither be created nor destroyed.



$$\dot{m}_{\text{cube}} = \dot{m}_{\text{in}} - \dot{m}_{\text{out}}$$

If more mass goes in than comes out, the mass of the container (cube) must change.

From the **Eulerian Point of View**,

$$\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = - \frac{\partial \rho}{\partial t}$$

Conservation of mass is defined as the sum of three spatial derivatives of a velocity field and set equal to the rate of change of mass in terms of density (for **compressible flow**) or set equal to zero (for **incompressible flow**).

$$\rho \left[\frac{\partial(u)}{\partial x} + \frac{\partial(v)}{\partial y} + \frac{\partial(w)}{\partial z} \right] = 0 \text{ (Incompressible flow)}$$

$\frac{\partial(\rho u)}{\partial x}$ = Represents how much different the flow rate is coming in than leaving in x – direction

If the fluid is leaving faster than coming in, the $(\partial(\rho u)/\partial x)$ term becomes **positive**.

$\frac{\partial(\rho v)}{\partial y}, \frac{\partial(\rho w)}{\partial z}$ = represents the fluid passing through y and z direction

$\frac{\partial \rho}{\partial t}$ = Change in mass of the cube (negative change in density over time)

$$\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = - \frac{\partial \rho}{\partial t}$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{V}) = 0$$

In the process, we make use of the following vector identity,

$$\nabla \cdot (\rho \vec{V}) \equiv \rho \nabla \cdot \vec{V} + \vec{V} \cdot \nabla \rho$$

Using the vector identity, the continuity equation becomes,

$$\frac{\partial \rho}{\partial t} + \vec{V} \cdot \nabla \rho + \rho \nabla \cdot \vec{V} = 0$$

The sum of the first two terms is the substantial (material) derivative of ρ . Hence,

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \vec{V} = 0$$

Momentum Equations (a.k.a Navier-Stokes Equation):

$$\rho \frac{Du}{Dt} = -\nabla P + \mu \nabla^2 u + \rho g$$

where,

$$\rho \frac{Du}{Dt} \Rightarrow \text{Mass} \times \text{Acceleration}$$

$\nabla P \Rightarrow$ Pressure force

$\mu \nabla^2 u \Rightarrow$ Viscous/Tangential force

$\rho g \Rightarrow$ Gravity/Body force

From Newton's second law,

$$F = ma$$

Linear momentum,

$$p = mV \Rightarrow \frac{dp}{dt} = m \frac{dV}{dt} = ma = F$$

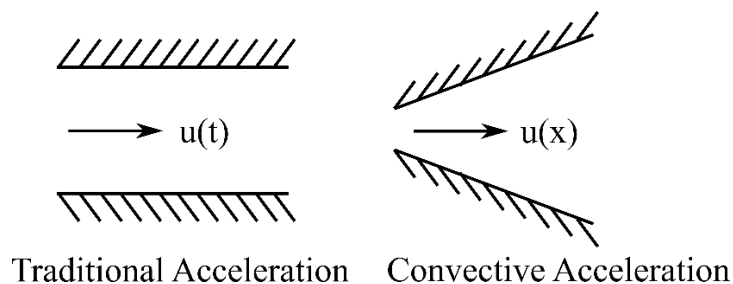
Force = Rate of change of momentum

In fluid dynamics,

The mass of the flow may not remain constant. Hence,

$$F = \frac{d(mV)}{dt} \text{ (for Lagrangian)}$$

From the **Eulerian Point of View**:



Mass times velocity per unit volume (R.H.S) can be written as,

$$\frac{D((m/\forall)V)}{Dt} = \frac{D(\rho V)}{Dt}$$

Where $\forall$ = **Volume**

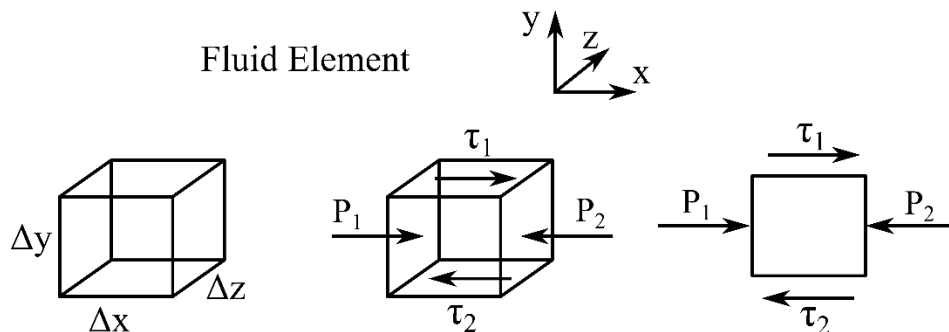
For an incompressible flow, density remain constant with time.

$$\frac{D(\rho V)}{Dt} = \rho \frac{DV}{Dt}$$

Forces (L.H.S)

Body forces: gravity, electromagnetic forces, or any other forces which “act at a distance” on the fluid inside V .

Surface forces: pressure and shear stress acting on the control surface S .



Force due to pressure per unit volume:

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}} \Rightarrow \text{Force} = \text{Pressure} \times \text{Area}$$

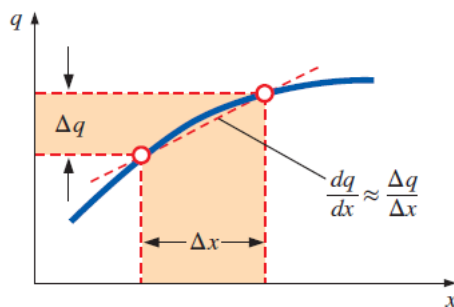
Force along x - direction due to pressure, $F_x = P_2 A_2 - P_1 A_1$

$$\text{Area} = A_1 = A_2 = \Delta y \Delta z$$

Force due to pressure = $\Delta P \times (\Delta y \Delta z)$

$$\text{Force per unit volume} = \frac{\Delta P \times (\Delta y \Delta z)}{\Delta x \Delta y \Delta z} = \frac{\Delta P}{\Delta x}$$

For small element, $\Delta \rightarrow 0$



A first-order finite difference approximation for derivative dq/dx is simply the change in dependent variable (q) divided by the change in independent variable (x).

$$\text{Force per unit volume due to pressure} = \frac{\partial P}{\partial x}$$

Force due to shear stress per unit volume:

$$\text{Stress} = \frac{\text{Force}}{\text{Area}} \Rightarrow \text{Force} = \text{Stress} \times \text{Area}$$

Force due to shear stress (tangential to surface), $\mathbf{F} = \tau_2 \mathbf{A}_2 - \tau_1 \mathbf{A}_1$

$$\text{Area} = \mathbf{A}_1 = \mathbf{A}_2 = \Delta x \Delta z$$

$$\text{Force due to shear stress} = \Delta \tau \times (\Delta x \Delta z)$$

$$\text{Force per unit volume} = \frac{\Delta \tau \times (\Delta x \Delta z)}{\Delta x \Delta y \Delta z} = \frac{\Delta \tau}{\Delta y}$$

We know that,

$$\text{Shear stress, } \tau = \text{Dynamic viscosity} \times \text{Velocity gradient} = \mu \frac{\Delta u}{\Delta y}$$

$$\text{Force per unit volume} = \frac{\mu \Delta (\Delta u / \Delta y)}{\Delta y} = \mu \frac{\Delta^2 u}{\Delta y^2}$$

For small element, $\Delta \rightarrow \partial$

$$\text{Force per unit volume} = \mu \frac{\partial^2 u}{\partial y^2}$$

In **3-D** flow along x – direction (molecules jump in all the three directions),

$$\text{Force per unit volume} = \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

Body Forces:

- Harder to physically interpret
- Magical forces acting on fluid particles without touching it.

Conservation of momentum generally depends on gravity force when we deal with fluid dynamics.

Force per unit volume:

$$\mathbf{F}_{\text{body}} = \frac{\mathbf{mg}}{\mathbf{V}} = \rho \mathbf{g}$$

The momentum equation along x, y, and z can be written as

$$\mathbf{x - direction} \rightarrow \rho \left[\frac{\partial \mathbf{u}}{\partial t} + u \frac{\partial \mathbf{u}}{\partial x} + v \frac{\partial \mathbf{u}}{\partial y} + w \frac{\partial \mathbf{u}}{\partial z} \right] = - \frac{\partial P}{\partial x} + \mu \left(\frac{\partial^2 \mathbf{u}}{\partial x^2} + \frac{\partial^2 \mathbf{u}}{\partial y^2} + \frac{\partial^2 \mathbf{u}}{\partial z^2} \right) + \rho a_x$$

$$\mathbf{y - direction} \rightarrow \rho \left[\frac{\partial \mathbf{v}}{\partial t} + u \frac{\partial \mathbf{v}}{\partial x} + v \frac{\partial \mathbf{v}}{\partial y} + w \frac{\partial \mathbf{v}}{\partial z} \right] = - \frac{\partial P}{\partial y} + \mu \left(\frac{\partial^2 \mathbf{v}}{\partial x^2} + \frac{\partial^2 \mathbf{v}}{\partial y^2} + \frac{\partial^2 \mathbf{v}}{\partial z^2} \right) + \rho a_y$$

$$\mathbf{z - direction} \rightarrow \rho \left[\frac{\partial \mathbf{w}}{\partial t} + u \frac{\partial \mathbf{w}}{\partial x} + v \frac{\partial \mathbf{w}}{\partial y} + w \frac{\partial \mathbf{w}}{\partial z} \right] = - \frac{\partial P}{\partial z} + \mu \left(\frac{\partial^2 \mathbf{w}}{\partial x^2} + \frac{\partial^2 \mathbf{w}}{\partial y^2} + \frac{\partial^2 \mathbf{w}}{\partial z^2} \right) + \rho a_z$$

Where,

$$\rho \left[\frac{\partial(\quad)}{\partial t} \right] = \text{Traditional Acceleration (function of time)}$$

$$\rho \left[u \frac{\partial(\quad)}{\partial x} + v \frac{\partial(\quad)}{\partial y} + w \frac{\partial(\quad)}{\partial z} \right] = \text{Convective Acceleration (function of position)}$$

$$\frac{\partial P}{\partial(\quad)} = \text{Pressure/Normal Surface Force}$$

$$\mu \left(\frac{\partial^2(\quad)}{\partial x^2} + \frac{\partial^2(\quad)}{\partial y^2} + \frac{\partial^2(\quad)}{\partial z^2} \right) = \text{Viscous/Tangential Force}$$

$$\rho a(\quad) = \text{Body/Gravity Force}$$

$$\text{x - direction} \rightarrow \frac{D(\rho u)}{Dt} + \rho u \left[\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right] = -\frac{\partial P}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) + \rho a_x$$

$$\text{y - direction} \rightarrow \frac{D(\rho v)}{Dt} + \rho v \left[\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right] = -\frac{\partial P}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) + \rho a_y$$

$$\text{z - direction} \rightarrow \frac{D(\rho w)}{Dt} + \rho w \left[\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right] = -\frac{\partial P}{\partial z} + \mu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) + \rho a_z$$

Common Assumptions:

1. Ignore the effect of body force(s) [Gas is very light]
2. The flow is inviscid ($\mu = 0$).
3. The flow is steady ($d(\quad)/dt$).

By taking these assumptions into consideration, we get **Euler's equation:**

- **No body forces, flow is inviscid and steady.**

$$\text{x - direction} \rightarrow u \frac{\partial(\rho u)}{\partial x} + v \frac{\partial(\rho u)}{\partial y} + w \frac{\partial(\rho u)}{\partial z} + \rho u \left[\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right] = -\frac{\partial P}{\partial x}$$

$$\text{y - direction} \rightarrow u \frac{\partial(\rho v)}{\partial x} + v \frac{\partial(\rho v)}{\partial y} + w \frac{\partial(\rho v)}{\partial z} + \rho v \left[\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right] = -\frac{\partial P}{\partial y}$$

$$\text{z - direction} \rightarrow u \frac{\partial(\rho w)}{\partial x} + v \frac{\partial(\rho w)}{\partial y} + w \frac{\partial(\rho w)}{\partial z} + \rho w \left[\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right] = -\frac{\partial P}{\partial z}$$

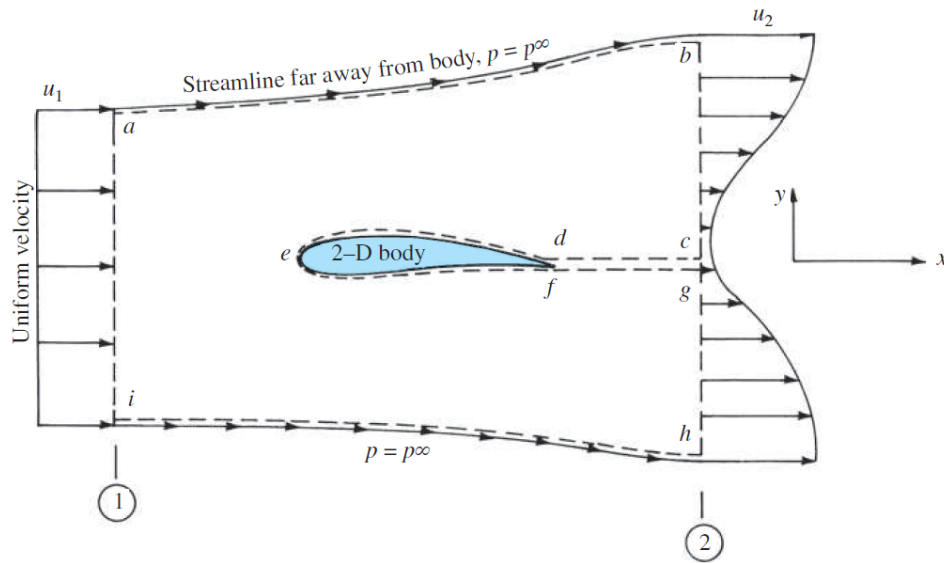
Bernoulli's Equation:

For an incompressible flow,

$$\rho \left[\frac{\partial(u)}{\partial x} + \frac{\partial(v)}{\partial y} + \frac{\partial(w)}{\partial z} \right] = 0 \text{ (Incompressible flow)}$$

$$P + \frac{1}{2} \rho V^2 = \text{Constant}$$

Momentum Deficit Method or Wake-Survey Method:



Assumptions:

1. Steady
2. Inviscid
3. No body forces
4. No change in the area
5. Constant pressure on all sides

Based on the assumptions, we get the momentum equation in the following form:

$$\iint_s (\rho \mathbf{V}_x \mathbf{ds}) \mathbf{V}_x = - \iint_s \mathbf{P} \mathbf{ds} - \mathbf{R}$$

$$\iint_s \mathbf{P} \mathbf{ds} = \text{Pressure distribution over the surface}$$

R = Resultant aerodynamic force due to pressure and shear stress on the airfoil

By Newton's third law, the body exerts equal and opposite resultant force on the flow (i.e. on the control surface bounded by def).

- Forces on cd and fg control surfaces cancels each other, because they are equal and opposite.

Surface force on def (control surface) created by the presence of body

$$\text{Surface force} = - \iint_s \mathbf{P} \mathbf{ds} - \mathbf{R}$$

Consider the unit depth ($ds = dy \cdot (1)$),

$$\int (\rho \mathbf{V}_x d\mathbf{y}) \mathbf{V}_x = - \int P d\mathbf{y} - \mathbf{R}$$

Constant pressure assumption:

$$\int (\rho \mathbf{V}_x d\mathbf{y}) \mathbf{V}_x = -\mathbf{R}$$

The x-component of R is aerodynamic drag per unit span.

$$\int (\rho \mathbf{V}_x d\mathbf{s}) \mathbf{V}_x = -\mathbf{D}$$

Momentum flux:

$$- \int_i^a (\rho u_1 d\mathbf{y}) u_1 + \int_b^h (\rho u_2 d\mathbf{y}) u_2 = -\mathbf{D}$$

From the conservation of mass:

$$\int_i^a \rho u_1 d\mathbf{y} = \int_h^b \rho u_2 d\mathbf{y}$$

Multiply by u_1 on both the sides, we get

$$\int_i^a (\rho u_1 d\mathbf{y}) u_1 = \int_h^b (\rho u_2 d\mathbf{y}) u_1$$

Substituting the above equation, we get

$$- \int_h^b (\rho u_2 d\mathbf{y}) u_1 + \int_b^h (\rho u_2 d\mathbf{y}) u_2 = -\mathbf{D}$$

Finally, we get

$$\rho \int_h^b u_2 (u_1 - u_2) d\mathbf{y} = \text{Drag}$$

By defining the velocity profile (u_2) through the **Gaussian distribution**, the **drag force** can be estimated.

Conservation of energy:

- This conservation law is rarely used and generally considered when the flow is **compressible** or encountering some **heating** or **cooling** effect.
- Energy can neither be created nor be destroyed.
- However, energy can change the form.
- Formulated from First law of thermodynamics.

$$\delta q + \delta w = \partial e$$

If there is a **work applied** to a system or **heat is added or leave** the system, the **internal energy** of the system changes.

The energy equation:

$$\frac{D}{Dt} \left[\rho \left(e + \frac{u^2}{2} \right) \right] + \rho \left(e + \frac{u^2}{2} \right) \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right) = \rho \dot{q} - \left(\frac{\partial(pu)}{\partial x} + \frac{\partial(pv)}{\partial y} + \frac{\partial(pw)}{\partial z} \right) + \rho (a_x u + a_y v + a_z w) + \text{Viscous terms}$$

Most of the terms in the energy equation have the same physical roots as the momentum equation. Because we are considering the work done by the forces instead of the forces.

$$\frac{D}{Dt} \left[\rho \left(e + \frac{u^2}{2} \right) \right] = \text{Traditional} + \text{Convective changes}$$

This term represents two forms of the energy of the fluid → Internal (e) + Kinetic energy.

$$\rho \left(e + \frac{u^2}{2} \right) \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right) = \text{Compression or Expansion term}$$

This term represents how energy leaves or enters the system as stuff expands out or compress in.

The right-hand side terms represent all the terms that can change the flows energy.

$$\rho \dot{q} = \text{Heat in or out of the system}$$

$$\left(\frac{\partial(pu)}{\partial x} + \frac{\partial(pv)}{\partial y} + \frac{\partial(pw)}{\partial z} \right) = \text{Workdone by pressure}$$

This term represents the pressure work in each direction.

$$\rho (a_x u + a_y v + a_z w) = \text{Workdone by body forces}$$

This term represents the work by the body force in each direction.

Common assumptions:

1. Steady ($d/dt = 0$)
2. Inviscid ($\mu = 0$)
3. Adiabatic ($\dot{q} = 0$)
4. No body forces, $a = 0$

Finally,

$$u \frac{\partial}{\partial x} \left[\rho \left(e + \frac{u^2}{2} \right) \right] + v \frac{\partial}{\partial y} \left[\rho \left(e + \frac{u^2}{2} \right) \right] + w \frac{\partial}{\partial z} \left[\rho \left(e + \frac{u^2}{2} \right) \right] = - \left(\frac{\partial(pu)}{\partial x} + \frac{\partial(pv)}{\partial y} + \frac{\partial(pw)}{\partial z} \right)$$

Note: Conservation of energy needs two more equations to get solved.

$$e = C_v T$$

$$P = \rho R T$$

Reference:

1. Anderson, J. (2011): *Fundamentals of Aerodynamics (SI units)*. McGraw hill.
2. Cengel, Y., & Cimbala, J. (2013): *Fluid mechanics fundamentals and applications (SI units)*. McGraw Hill.